

day 5, session 6, correlation and regression analysis

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setting a working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(dplyr)
library(tidyverse)
library(ggplot2)
library(rio)
```

import file

```
npk <- read.csv("gsssubset.csv")
head(npk)
```

	id	sex	degree	income	marital	age	height	weight	hrswrk
1	1	MALE	BACHELOR	60967.50	DIVORCED	53	72	190	60
2	2	FEMALE	BACHELOR	60967.50	MARRIED	26	60	97	40
3	4	FEMALE	BACHELOR	10161.25	MARRIED	56	68	160	20
4	14	FEMALE	HIGH SCHOOL	17551.25	MARRIED	40	65	156	37
5	16	MALE	HIGH SCHOOL	17551.25	MARRIED	56	NA	210	6
6	19	MALE LT	HIGH SCHOOL	15703.75	MARRIED	51	68	170	NA

correlation and regression analysis

defn: correlation; measure the strength and direction of linear relationship b/w 2 var

correlation analysis

```
dataset <- data.frame(x = c(1:5), y = c(2, 4, 5, 7, 6))
head(dataset)
```

	x	y
1	1	2
2	2	4
3	3	5
4	4	7
5	5	6

pearson correlation

```
pearson_corr <- cor.test(dataset$x, dataset$y, method = "pearson")
print(paste("pearson correlation:", pearson_corr))
```

```
[1] "pearson correlation: c(t = 3.6666666666666666)"
[2] "pearson correlation: c(df = 3)"
[3] "pearson correlation: 0.035081514715482"
[4] "pearson correlation: c(cor = 0.904194430179465)"
[5] "pearson correlation: c(correlation = 0)"
[6] "pearson correlation: two.sided"
[7] "pearson correlation: Pearson's product-moment correlation"
[8] "pearson correlation: dataset$x and dataset$y"
[9] "pearson correlation: c(0.108415141575197, 0.993725716879883)"
```

There is strong correlation b/n x and y
is association significant?
test significance of pearson_correlation

H0:

H1:

```
pearson_corr_test <- cor.test(dataset$x, dataset$y, method = "pearson")  
print(pearson_corr_test)
```

```
Pearson's product-moment correlation  
  
data: dataset$x and dataset$y  
t = 3.6667, df = 3, p-value = 0.03508  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
 0.1084151 0.9937257  
sample estimates:  
      cor  
0.9041944
```

- since the p-value < 5%, we reject the null hypothesis and conclude that there exists a relation that is significant b/n x and y

spearman correlation

```
spearman_corr <- cor.test(dataset$x, dataset$y, method = "spearman")  
print(paste("spearman correlation:", pearson_corr))
```

```
[1] "spearman correlation: c(t = 3.66666666666666)"  
[2] "spearman correlation: c(df = 3)"  
[3] "spearman correlation: 0.035081514715482"  
[4] "spearman correlation: c(cor = 0.904194430179465)"  
[5] "spearman correlation: c(correlation = 0)"  
[6] "spearman correlation: two.sided"  
[7] "spearman correlation: Pearson's product-moment correlation"  
[8] "spearman correlation: dataset$x and dataset$y"  
[9] "spearman correlation: c(0.108415141575197, 0.993725716879883)"
```

strong positive correlation
is the relationship significant?

```
spearman_corr_test <- cor.test(dataset$x, dataset$y, method = "spearman")  
print(spearman_corr_test)
```

```
Spearman's rank correlation rho  
  
data: dataset$x and dataset$y  
S = 2, p-value = 0.08333  
alternative hypothesis: true rho is not equal to 0  
sample estimates:  
rho  
0.9
```

- since the p-value > 5%, we fail to reject the null hypothesis and conclude that there is no significant difference between x and y variables

regression analysis

defn: regression; rltshp btn independent \$ dependent variables

simple linear regration

```
model <- lm(y ~ x, data = dataset)
summary(model)
```

```
Call:
lm(formula = y ~ x, data = dataset)

Residuals:
    1     2     3     4     5 
-0.6  0.3  0.2  1.1 -1.0 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.500      0.995   1.508   0.2288
x              1.100      0.300   3.667   0.0351 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9487 on 3 degrees of freedom
Multiple R-squared:  0.8176,    Adjusted R-squared:  0.7568 
F-statistic: 13.44 on 1 and 3 DF,  p-value: 0.03508
```

- a unit increase in variable x increses variable y by 1.1unit, and effect is significant since the p-value <5%
- var X, explains variabvle y by 81,76% but the remainingpercentage is explained by other factors not in our model

read data series

muultiple linear regression

```
data1 <- data.frame(y = c(2, 4, 5, 4, 6), x1 = c(1:5), x2 = c(3, 5, 7, 6, 8))
modell <- lm(y ~ x1 + x2, data = data1)
summary(modell)
```

```
Call:
lm(formula = y ~ x1 + x2, data = data1)

Residuals:
    1         2         3         4         5 
-0.06667  0.33333 -0.26667 -0.20000  0.20000 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.4222    0.6748  -0.626   0.5954
x1            -0.1778    0.2703  -0.658   0.5784
x2             0.8889    0.2222   4.000   0.0572 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3651 on 2 degrees of freedom
Multiple R-squared:  0.9697,    Adjusted R-squared:  0.9394 
F-statistic: 32 on 2 and 2 DF,  p-value: 0.0303
```

- a unit increase in x1, decreases y by 0.1778
- a unit increase in x2, increases y by 0.8889
- var X1 and x2, explains variabvle y by 0.9697 but the remainingpercentage is explained by other factors not in our model

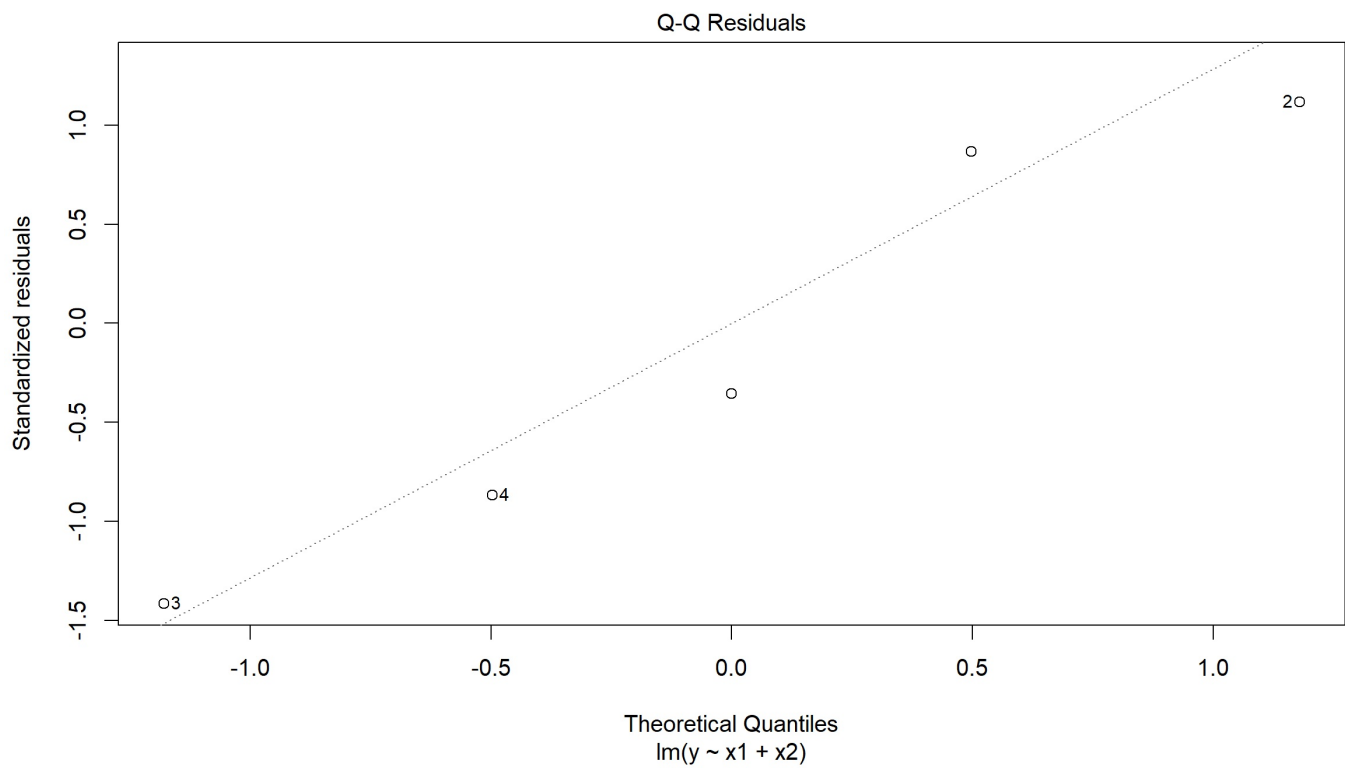
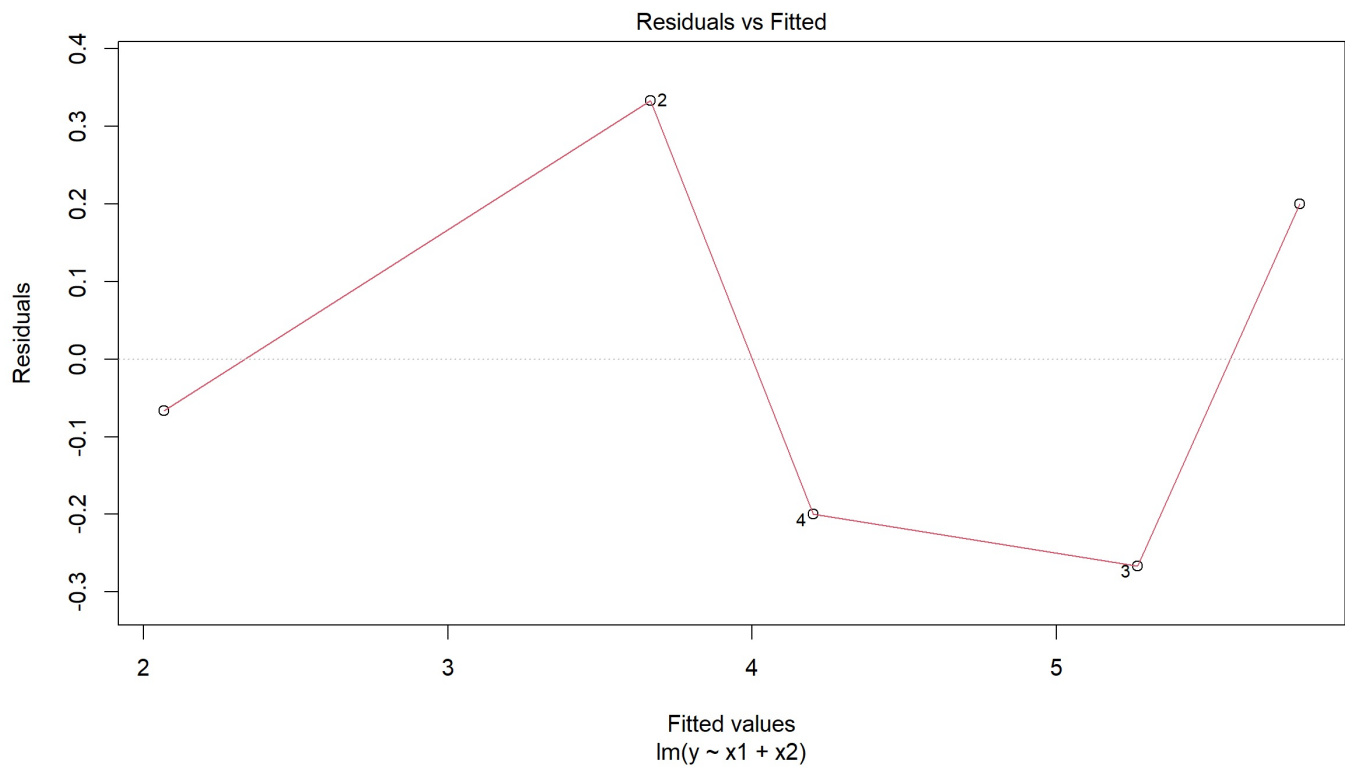
check whether modell is significance

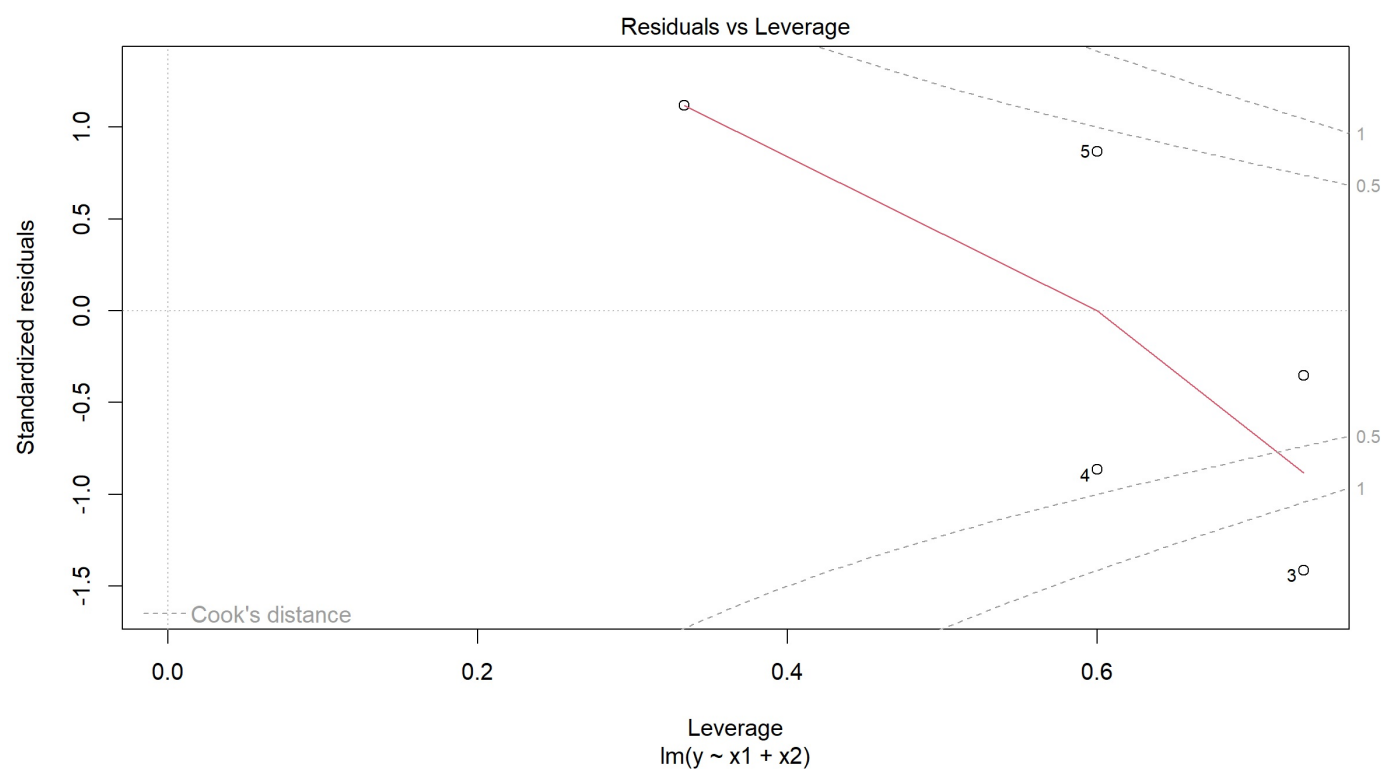
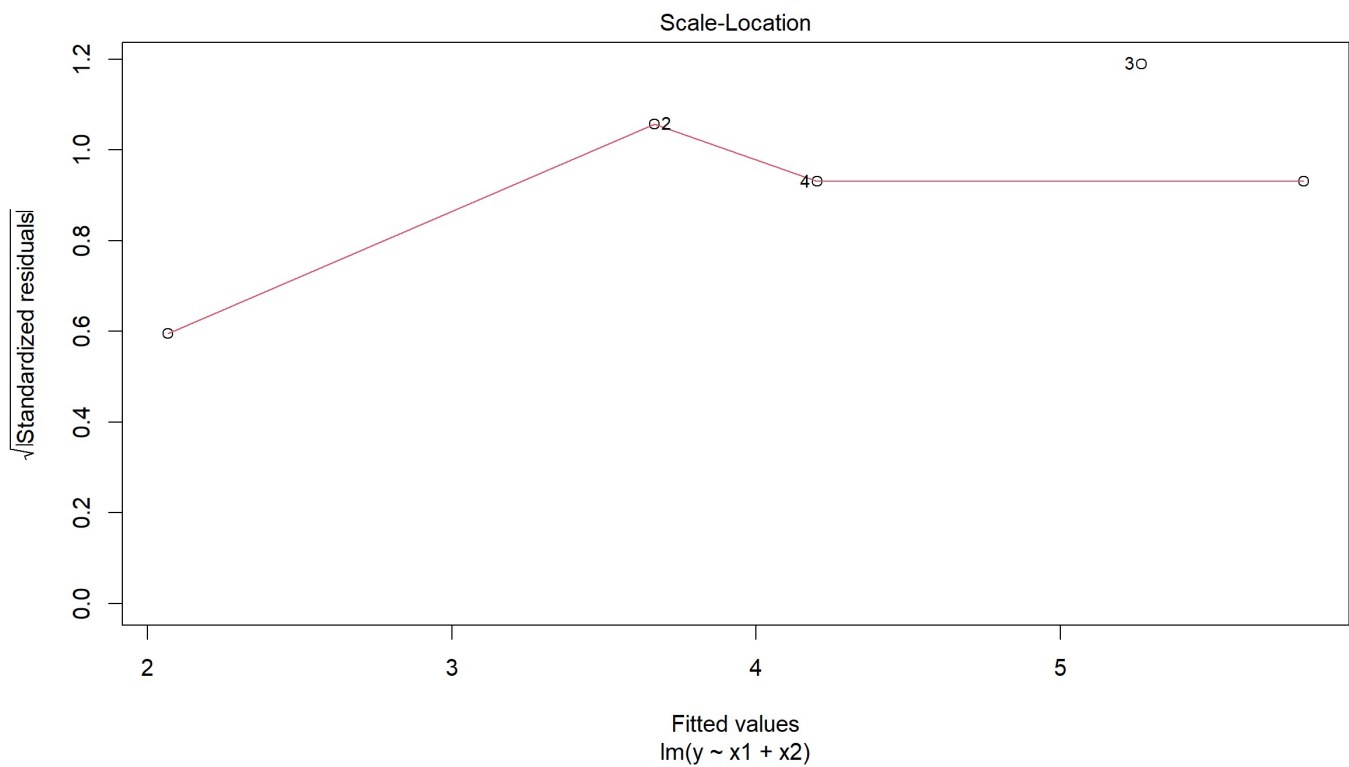
model diagnosis

```
par(mfrow <- c(1, 2))
```

```
NULL
```

```
plot(model1)
```





import data set

```
nck <- read.csv("gsssubset.csv")
head(nck)
```

	id	sex	degree	income	marital	age	height	weight	hrswrk
1	1	MALE	BACHELOR	60967.50	DIVORCED	53	72	190	60
2	2	FEMALE	BACHELOR	60967.50	MARRIED	26	60	97	40
3	4	FEMALE	BACHELOR	10161.25	MARRIED	56	68	160	20
4	14	FEMALE	HIGH SCHOOL	17551.25	MARRIED	40	65	156	37
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6	19	MALE LT	HIGH SCHOOL	15703.75	MARRIED	51	68	170	NA

Calculate pearson correlartion

```
pearson_test <- cor.test(nck$income, nck$hrswrk, method = "pearson")
print(paste("pearson correlation:", pearson_test))
```

```
[1] "pearson correlation: c(t = 11.4631959502933)"
[2] "pearson correlation: c(df = 980)"
[3] "pearson correlation: 1.23016283434493e-28"
[4] "pearson correlation: c(cor = 0.343850428740101)"
[5] "pearson correlation: c(correlation = 0)"
[6] "pearson correlation: two.sided"
[7] "pearson correlation: Pearson's product-moment correlation"
[8] "pearson correlation: nck$income and nck$hrswrk"
[9] "pearson correlation: c(0.287475367853873, 0.397851204850185)"
```

testing significance

```
pearson_test_sign <- cor.test(nck$income, nck$hrswrk, method = "pearson")
print(pearson_test_sign)
```

Pearson's product-moment correlation

data: nck\$income and nck\$hrswrk
t = 11.463, df = 980, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
0.2874754 0.3978512
sample estimates:
cor
0.3438504

- since the p-value<5%, we reject null analysis and conclude that there exist significance between income and hours of work

select continous var

```
nck_selected <- nck %>%
  select(hrswrk, age, weight, income, height)
cor(nck_selected)
```

	hrswrk	age	weight	income	height
hrswrk	1	NA	NA	NA	NA
age	NA	1	NA	NA	NA
weight	NA	NA	1	NA	NA
income	NA	NA	NA	1	NA
height	NA	NA	NA	NA	1

a correlation matrix

test significance

testing significance

regression analysis

```
model2 <- lm(income ~ age + height + weight + hrswrk, data = nck)
summary(model2)
```

Call:
lm(formula = income ~ age + height + weight + hrswrk, data = nck)

Residuals:

	Min	1Q	Median	3Q	Max
	-72623	-18221	-6209	9623	127699

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.320e+05	1.768e+04	-7.467	1.84e-13	***
age	5.582e+02	7.746e+01	7.206	1.17e-12	***
height	1.665e+03	2.888e+02	5.767	1.09e-08	***
weight	2.556e+00	2.797e+01	0.091	0.927	
hrswrk	7.370e+02	7.257e+01	10.156	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31430 on 956 degrees of freedom
(33 observations deleted due to missingness)
Multiple R-squared: 0.2016, Adjusted R-squared: 0.1982
F-statistic: 60.34 on 4 and 956 DF, p-value: < 2.2e-16